.NET TECHNOLOGIES
(01CE1523)
Lab Manual

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Lab	**A**

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Lab	Program	CO	Date	Marks				Total	Signature
				R1(4)	R2(2)	R3(2)	R4(2)	(10)	
1.	Develop Student Admission Management Module using OOP Concepts (Class, Object, Constructor, Access Modifiers).	CO1							
2.	Design Employee Payroll System using Inheritance, Interface & Polymorphism	CO1							
3.	Develop Expense Tracking Module with Exception Handling	CO1							
4.	Build Online Event Registration Portal using ASP.NET Controls & Validation	CO2							
5.	Develop Academic Calendar & Leave Management System using Rich Controls, Session & Cookies	CO2							
6.	Create Product Catalog Application using ASP.NET MVC Architecture. (Controllers, Models, Views, Routing)	CO3							
7.	Develop Feedback Management Module using MVC Layout, Partial View & Validation. (HTML Helpers, Validation, Layouts)	CO3							
8.	Develop Inventory Management Module using ADO.NET & CRUD Operations	CO4							
9.	Create Placement	CO4							

	Dashboard using Data Controls & Crystal Reports								
10.	Build Student Management REST API using ASP.NET Core (Middleware, API, Project Structure)	CO5							
11.	Develop Smart Campus Service Portal – Feature Extension (Integrate MVC + Database + API)	CO1, CO2, CO3, CO4, CO5							

R1	Practical Performance
R2	Timely Submission
R3	Neatness & Documentation
R4	Uploaded on Git

Experiment 6

Create Product Catalog Applicationusing ASP.NET MVC Architecture. (Controllers, Models, Views, Routing)

RouteConfig.cs

```
usingSystem;
usingSystem.Collections.Generic;
usingSystem.Linq;
usingSystem.Web;
usingSystem.Web.Mvc;
usingSystem.Web.Routing;

namespace practical_06_.NET
{
   publicclass RouteConfig
   {
      public static void RegisterRoutes(RouteCollection routes)
      {
         routes.IgnoreRoute("{resource}.axd/{*pathInfo}");

         routes.MapRoute(
            name: "Default",
            url: "{controller}/{action}/{id}",
            defaults: new { controller = "Home", action = "Index", id = UrlParameter.Optional }
         );
      }
   }
}
```

ProductController.cs

```
usingSystem.Collections.Generic;
usingSystem.Web.Mvc;
usingpractical_06_.NET.Models;

namespace practical_06_.NET.Controllers
{
   public class ProductController : Controller
   {

      public ActionResult Index()
```

```
    {
        List<Product> products = new List<Product>()
        {
            new Product
            {
                Id = 1,
                Name = "Laptop",
                Price = 55000,
                Category = "Electronics"
            },

            new Product
            {
                Id = 2,
                Name = "Mouse",
                Price = 800,
                Category = "Accessories"
            },

            new Product
            {
                Id = 3,
                Name = "Keyboard",
                Price = 1500,
                Category = "Accessories"
            },

            new Product
            {
                Id = 4,
                Name = "Headphones",
                Price = 2500,
                Category = "Audio"
            }
        };

        return View(products);
    }
  }
}
```

Product.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
```

```
using System.Web;


namespace practical_06_.NET.Models
{
    public class Product
    {
        public int Id { get; set; }

        public string Name { get; set; }

        public decimal Price { get; set; }

        public string Category { get; set; }
    }
}
```

Indes.cshtml

```
@model IEnumerable<practical_06_.NET.Models.Product>

@{
    Layout = null;
}

<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <title>Product Catalog</title>

    <style>
        * {
            box-sizing: border-box;
        }

        body {
            margin: 0;
            font-family: Arial, Helvetica, sans-serif;
            background: linear-gradient(135deg, #667eea, #764ba2);
            min-height: 100vh;
            padding: 50px 20px;
        }

        .container {
```

```
    width: 90%;
    max-width: 1100px;
    margin: auto;
    background: white;
    padding: 35px;
    border-radius: 18px;
    box-shadow: 0 15px 40px rgba(0, 0, 0, 0.25);
}

.header {
    text-align: center;
    margin-bottom: 30px;
}

    .header h1 {
        margin: 0;
        color: #333;
        font-size: 36px;
    }

    .header p {
        color: #777;
        margin-top: 8px;
        font-size: 16px;
    }

table {
    width: 100%;
    border-collapse: collapse;
    overflow: hidden;
    border-radius: 12px;
}

th{
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    padding: 16px;
    font-size: 16px;
    text-align: center;
}

td{
    padding: 15px;
    text-align: center;
    border-bottom: 1px solid #eee;
```

```
      color: #444;
      font-size: 15px;
    }

    tbody tr:nth-child(even) {
      background-color: #f7f7ff;
    }

    tbody tr:hover {
      background-color: #e9e7ff;
      transform: scale(1.01);
      transition: 0.2s;
    }

    .price {
      color: #198754;
      font-weight: bold;
    }

    .category {
      display: inline-block;
      padding: 6px 12px;
      border-radius: 20px;
      background-color: #eee8ff;
      color: #6741d9;
      font-weight: bold;
      font-size: 13px;
    }

    .footer {
      text-align: center;
      margin-top: 25px;
      color: #888;
      font-size: 14px;
    }
  </style>
</head>

<body>

  <div class="container">

    <div class="header">
      <h1> Product Catalog</h1>
      <p>Explore our collection of products</p>
```

```
</div>

<table>

  <thead>
    <tr>
      <th>ID</th>
      <th>Product Name</th>
      <th>Price</th>
      <th>Category</th>
  </tr>
  </thead>

  <tbody>

    @foreach(var product in Model)
    {
      <tr>
        <td>@product.Id</td>

        <td>
          <strong>@product.Name</strong>
        </td>

        <tdclass="price">
          ₹@product.Price
        </td>

        <td>
          <span class="category">
            @product.Category
          </span>
        </td>
      </tr>
    }

  </tbody>

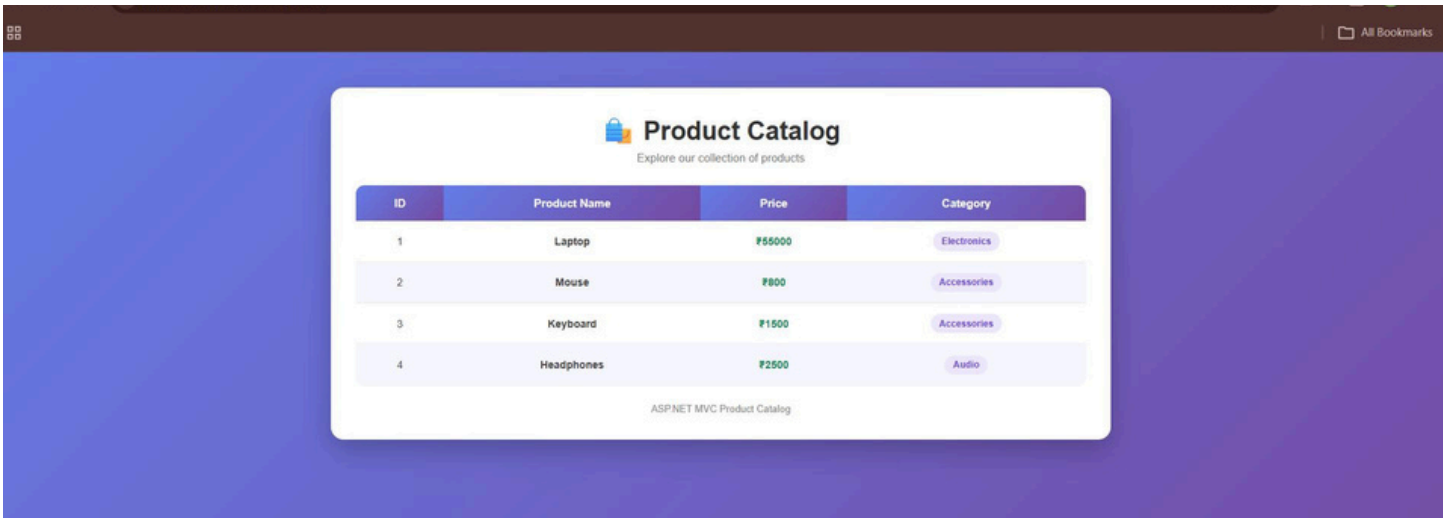
</table>

<divclass="footer">
  ASP.NETMVC Product Catalog
</div>

</div>
```

```
</body>
</html>
```

OUTPUT SCREENSHOT



GitHubURL: _____